THE $N^{\sqrt{2}+o(1)}$ CONSTRUCTION FOR HURWITZ'S PROBLEM, WRITTEN OUT IN TERMS OF BILINEAR FORMS

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ABSTRACT. For every N there are integer bilinear forms $f_1, g_1, \dots, f_K, g_K$ in $x_1, \dots, x_N, y_1, \dots, y_N$ with $h!\,(\sum_i x_i^2)(\sum_j y_j^2) = \sum_{k\le K} f_k g_k$ and $K = N^{\sqrt{2}+o(1)}$, where $h = O(\log^2 N)$ is the least integer with at least N partitions. This is a consequence of Theorem B of the paper *A $2^{o(n)}$ time algorithm for 3-SAT* [1], first pointed out in [8]. This note restates the construction directly as an identity of bilinear forms, gives the coefficients explicitly, writes out the proofs of the two lemmas it rests on with the compressed steps filled in, and says exactly what is machine-checked. Nothing here is new mathematics beyond [1]; the point is to make the construction checkable by a reader who works with polynomials rather than with the function language of [1].

1. The result and its status

Write $p(E)$ for the number of partitions of E ($p(0) = 1$, $p(E) = 0$ for $E < 0$), $\mathcal{P}_h$ for the set of partitions of h, and $B_F(N)$ for the least s with $(\sum_{i\le N} x_i^2)(\sum_{j\le N} y_j^2) = \sum_{k\le s} f_k f_k'$, f_k, f_k' bilinear over F (the quantity $B_F(\mathrm{SOS}_N)$ of [6], and the bilinear-product variant (9) of [4]).

Theorem 1. *Let $h \ge 1$ and $K_h = \sum_{j=0}^{2h} (4h-2j+1)\,p(j)$. There are bilinear forms f_k, g_k ($k \le K_h$; some products $f_k g_k$ may vanish) in variables x_ξ, y_η indexed by $\xi, \eta \in \mathcal{P}_h$, with integer coefficients given explicitly in Section 3, such that*

$$h!\Big(\sum_{\xi\in\mathcal{P}_h} x_\xi^2\Big)\Big(\sum_{\eta\in\mathcal{P}_h} y_\eta^2\Big) = \sum_{k=1}^{K_h} f_k g_k.$$

Setting all but $N \le p(h)$ of the x's and y's to zero gives the identity for N variables. Taking h least with $p(h) \ge N$, the Hardy–Ramanujan asymptotics give $K_h = N^{\sqrt{2}+o(1)}$ (Section 5, for $N = 2^\lambda$; for other N use $2^{\lambda-1} < N \le 2^\lambda$). Hence:

Corollary 2. *$B_{\mathbb{Q}}(N) \le N^{\sqrt{2}+o(1)}$, and the same over any field of characteristic 0 or larger than $h = O(\log^2 N)$. By* [6, Remark 1.5]*, $\sigma_{\mathbb{C}}(N) \le 3B_{\mathbb{C}}(N) \le N^{\sqrt{2}+o(1)}$, where $\sigma_{\mathbb{C}}(N)$ is the least number of squares in Hurwitz's problem over $\mathbb{C}$.*

The previous best upper bound is $N^{c+o(1)}$ with $c = 1 + 2H(\alpha/2) \approx 1.6147$, $H(\alpha) = \frac{1}{2}$ ([4, Theorem 1 and its proof in §4], stated there as $O(N^{c'})$ with $c' < 1.62$; for the bilinear-product form (9) see [4, §5.1, Remark 18], and [5, Theorem 12] for a direct proof over every field). It also holds in positive characteristic, where the present identity says nothing ($h! = 0$ once $p \le h$). The open problem in [4, §7] asks whether $\sigma_{\mathbb{C}}(N) \le N^{1.5+o(1)}$ is attainable by parity representations; the bound here is below that target, by a different route.

Remark 3 (a consequence we have not checked beyond substitution). The proof of [4, Theorem 25] uses Theorem 1 there only as a black box, $\sigma(n) = O(n^{1+\delta})$, and obtains $\mathrm{size}^{(nc)}(\mathrm{ID}_n) \le n^{(3+\omega(3)+\delta)/4+o(1)}$. Substituting $\delta = \sqrt{2}-1$ and $\omega(3) < 4.1997$ gives exponent below 1.904 (instead of 1.954), over fields of characteristic 0.

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Status. Theorem B of [1] is formalized in Lean 4 with Mathlib in [2] (commit `16a5ce41`): the statement file `eth/proof/statements/AppHalfPiecesStatement.lean` restates Appendix B in the paper's words (definitions and statements only, about 240 lines, importing Mathlib and two short statement files), and `GeneralPieces.thmHalfPiecesPure` proves it. We rebuilt this from source (`lake exe cache get; lake build GeneralPieces.Main`) and checked `#print axioms`: only `propext`, `Classical.choice`, `Quot.sound`. We then stated Corollary 2 (the bound over $\mathbb{Q}$, for N a power of two) in terms of polynomials, in a file that imports Mathlib only (excerpt, inside `namespace HurwitzSqrt2` with `open MvPolynomial Finset`; docstrings omitted):

```
abbrev Idx (lam : ℕ) : Type := Fin lam → Bool
abbrev Var (lam : ℕ) : Type := Idx lam ⊕ Idx lam
noncomputable def bil {lam : ℕ} (F : Idx lam → Idx lam → ℚ) : MvPolynomial (Var lam) ℚ :=
  Σ i, Σ j, C (F i j) * X (Sum.inl i) * X (Sum.inr j)
noncomputable def sos (lam : ℕ) : MvPolynomial (Var lam) ℚ :=
  (Σ i, X (Sum.inl i) ^ 2) * (Σ j, X (Sum.inr j) ^ 2)
def Statement : Prop :=
  ∃ c : ℕ, ∀ lam : ℕ, 1 ≤ lam →
    ∃ (K : ℕ) (f g : Fin K → Idx lam → Idx lam → ℚ),
      (K : ℝ) ≤ (2 * lam : ℝ) ^ c * 2 ^ (Real.sqrt 2 * lam) ∧
      sos lam = Σ k, bil (f k) * bil (g k)
```

and proved `Statement` from `thmHalfPiecesPure` (file `HurwitzProof.lean`, 292 lines). That file restates the `HurwitzSqrt2` block rather than importing it; we diffed the two blocks and they are identical, and `#print axioms hurwitzSqrt2` gives the same three axioms. The Lean statement has rational coefficients and does not contain the factor $h!$, integrality, or the exact count K_h (its K is $\sum_s 2^{\lceil \log_2 \min(R_s^{\mathrm{row}}, R_s^{\mathrm{col}}) \rceil} \le 2K_h$); those are checked numerically for $h \le 5$ (Section 6) and follow from (54), (55). So the claim rests on Lean's kernel plus a reading of the definitions above and of Definition 7 as formalized (`Pieces.Family.Is` in `PiecesStatement.lean`, which matches the printed definition field by field, except the running-time clause of 7(c), which the bound does not need). The Lean development was written and reviewed by AI models; no person is known to have read its proofs. Section 6 maps each step below to its Lean declaration.

2. The construction

2.1. **The Fock space.** Let $V = \mathbb{Q}[t_1, t_2, \dots]$, with t_k of weight k; the monomials $t^\mu = \prod_i t_{\mu_i}$, μ a partition, form a basis, and V_E (weight E) has dimension $p(E)$. For $k \ge 1$ let

$$\nabla_{-k} = t_k \cdot, \qquad \nabla_k = k\, \partial/\partial t_k, \qquad \text{so} \quad [\nabla_k, \nabla_{-l}] = k\, \delta_{kl}.$$

These are the Heisenberg relations; V is the Fock representation. ∇_k maps $V_E \to V_{E-k}$ for all $k \ne 0$. For σ, σ' partitions and an operator T, write $[t^{\sigma'}]\, T\, t^\sigma$ for the matrix entry.

2.2. **The operators $a[r]$.** For a monomial $a = t^\mu$ with parts $\mu_1, \dots, \mu_n$ (listed with multiplicity) and $r \in \mathbb{Z}$, put

$$a[r] = \sum_{\substack{k_1, \dots, k_n \in \mathbb{Z} \setminus \{0\} \\ k_1 + \cdots + k_n = -r}} \prod_i \binom{-k_i - 1}{\mu_i - 1} \prod_{i : k_i < 0} \nabla_{k_i} \prod_{i : k_i > 0} \nabla_{k_i}, \tag{47}$$

with all differentiations ($k_i > 0$) applied first, and extend linearly in a. Here $\binom{k}{j} = k(k-1)\cdots(k-j+1)/j!$ for all integers k. On V_E only finitely many terms are nonzero (the positive k_i add up to at most E), and $a[r]$ maps V_E into V_{E+r}, with integer matrix entries if a has integer coefficients. For example $t_1[r] = \nabla_{-r}$ ($r \ne 0$). With $Y(a, \omega) = \sum_r a[r]\, \omega^{r - |\mu|}$, these are the coefficients of the fields of the Heisenberg vertex algebra [3, §2.3, (4)], as [1, App. H] notes; we do not use this except in Remark 6.

2.3. **The operators Q_s.** Let $h \geq 1$, $b \in V_h$, and $a = \sum_{h'=0}^{h} \bar{a}_{h'}$ with $\bar{a}_{h'} \in V_{h'}$. Put $a^{\langle i \rangle} = \sum_{h'} \binom{h'+h}{i} \bar{a}_{h'}$ and, for $s \in \mathbb{Z}$,

$$Q_s(a,b) = \sum_{i=0}^{2h} (-1)^i \, a^{\langle i \rangle}[s-h+i] \; b[h-s-i], \qquad Q'_s(a,b) = \sum_{i=0}^{2h} (-1)^i \, b[h-s-i] \; a^{\langle i \rangle}[s-h+i]. \tag{48}$$

Both preserve weight. The weighting $a^{\langle i \rangle}$ makes Q_s linear in a while each homogeneous component $\bar{a}_{h'}$ is paired with its own binomial row $\binom{h'+h}{i}$; this is what lets the proof below handle inhomogeneous a one component at a time.

Lemma 4 (Lemma 9 of [1])**.** *On V_E, for every $E \geq 0$: (a) $Q_s(a,b) = Q'_s(a,b)$; (b) $Q_s(a,b) = 0$ for $|s| > E + h$; (c) $\sum_{|s| \leq E+h} Q_s(a,b) = (a[-h]b) \cdot \mathrm{Id}$, where $a[-h]b \in V_0 = \mathbb{Q}$.*

2.4. **Dual families.** For $n \geq 1$ let $y_n = \sum_{j=1}^{n} \binom{n}{j} t_j$ (no relation to the variables y_η of Theorem 1), and for $\mu \in \mathcal{P}_h$

$$a^\mu = \prod_i (-1)^{\mu_i - 1} y_{\mu_i}, \qquad b^\mu = \frac{h!}{z_\mu} \, t^\mu, \qquad z_\mu = \prod_k k^{m_k(\mu)} m_k(\mu)!,$$

where $m_k(\mu)$ is the number of parts equal to k; $h!/z_\mu \in \mathbb{Z}$ is the number of permutations of cycle type μ. a^μ has integer coefficients and only monomials of weight $\leq h$.

Lemma 5 (Lemma 10 of [1])**.** *$a^\mu[-h]\, b^\nu = h!\, \delta_{\mu\nu}$ for all $\mu, \nu \in \mathcal{P}_h$.*

2.5. **The identity.** For $\xi, \eta, \upsilon, \nu \in \mathcal{P}_h$ and $|s| \leq 2h$ let

$$A_s(\xi, \eta, \upsilon, \nu) = [t^\upsilon] \; Q_s(a^\xi, b^\nu) \; t^\eta. \tag{53}$$

By Lemma 4(b),(c) with $E = h$ and Lemma 5,

$$\sum_{|s| \leq 2h} A_s(\xi, \eta, \upsilon, \nu) = h! \, \delta_{\xi\nu} \, \delta_{\eta\upsilon}. \tag{1}$$

Equivalently, the degree-four polynomial $\sum_s \sum A_s(\xi, \eta, \upsilon, \nu) \, x_\xi y_\eta x_\nu y_\upsilon$ equals $h!(\sum x_\xi^2)(\sum y_\eta^2)$, already at the level of coefficient tensors (so also for the non-commutative ID polynomial of [6], with $(\xi, \eta, \nu, \upsilon)$ in the positions X, Y, X, Y). What remains is to write each tensor A_s as a short sum of products of two matrices, splitting the indices as $(\xi, \upsilon) \,|\, (\nu, \eta)$ or as $(\xi, \eta) \,|\, (\nu, \upsilon)$; commutatively, each such product is a product of two bilinear forms. (These two splittings are forms (b) and (a) of [6, Lemma 3.5], which matters for the non-commutative width of $h!\,\mathrm{ID}_N$; Theorem 1 itself does not need that lemma.)

3. The bilinear forms

Define integer matrices (rows and columns indexed by partitions) for each $\xi, \nu \in \mathcal{P}_h$:

$$\alpha_r^{(i)}[\xi]_{\sigma'\sigma} = (-1)^i \, [t^{\sigma'}] \; (a^\xi)^{\langle i \rangle}[r] \; t^\sigma, \qquad \beta_r[\nu]_{\sigma'\sigma} = [t^{\sigma'}] \; b^\nu[r] \; t^\sigma.$$

Inserting $\mathrm{Id} = \sum_\sigma t^\sigma [t^\sigma]$ on the middle weight space in (48) gives two factorizations.

Column-wise (directly from Q_s; display (55) of [1]): $b[h - s - i] \, t^\eta \in V_{2h-s-i}$, so

$$A_s(\xi, \eta, \upsilon, \nu) = \sum_{i=0}^{2h} \sum_{\sigma \in \mathcal{P}_{2h-s-i}} \alpha_{s-h+i}^{(i)}[\xi]_{\upsilon\sigma} \; \beta_{h-s-i}[\nu]_{\sigma\eta},$$

and each term (i, σ) gives the pair

$$f_{s,i,\sigma} = \sum_{\xi, \upsilon} \alpha_{s-h+i}^{(i)}[\xi]_{\upsilon\sigma} \, x_\xi y_\upsilon, \qquad g_{s,i,\sigma} = \sum_{\nu, \eta} \beta_{h-s-i}[\nu]_{\sigma\eta} \, x_\nu y_\eta.$$

Row-wise (from Q'_s, so it uses Lemma 4(a); display (54)): $a^{\langle i\rangle}[s-h+i]\,t^\eta \in V_{s+i}$, so

$$A_s(\xi,\eta,\upsilon,\nu)=\sum_{i=0}^{2h}\sum_{\sigma\in\mathcal{P}_{s+i}} \alpha^{(i)}_{s-h+i}[\xi]_{\sigma\eta}\,\beta_{h-s-i}[\nu]_{\upsilon\sigma},$$

$$f_{s,i,\sigma}=\sum_{\xi,\eta}\alpha^{(i)}_{s-h+i}[\xi]_{\sigma\eta}\,x_\xi y_\eta,\qquad g_{s,i,\sigma}=\sum_{\nu,\upsilon}\beta_{h-s-i}[\nu]_{\upsilon\sigma}\,x_\nu y_\upsilon.$$

In both cases $f_{s,i,\sigma}g_{s,i,\sigma}$, summed over the terms, has coefficient $A_s(\xi,\eta,\upsilon,\nu)$ on $x_\xi y_\eta x_\nu y_\upsilon$, so by (1) the sum over s is $h!(\sum x^2)(\sum y^2)$.
Which way for which s. The numbers of terms are $R^{\mathrm{row}}_s=\sum_{i=0}^{2h}p(s+i)$ and $R^{\mathrm{col}}_s=\sum_{i=0}^{2h}p(2h-s-i)$. Substituting $i\mapsto 2h-i$ shows $R^{\mathrm{col}}_s=R^{\mathrm{row}}_{-s}$, and R^{row}_s is nondecreasing in s. So the column-wise form is at least as short for $s\ge 0$ and the row-wise form for $s\le 0$, and

$$\min(R^{\mathrm{row}}_s,R^{\mathrm{col}}_s)=\sum_{j=0}^{2h-|s|}p(j),\qquad K_h=\sum_{|s|\le 2h}\sum_{j=0}^{2h-|s|}p(j)=\sum_{j=0}^{2h}(4h-2j+1)\,p(j).$$

This proves Theorem 1, given Lemmas 4 and 5. Note that Lemma 4(a) is used only for $s<0$. The intuition: both factors of a term pass through one weight space, of weight $2h-s-i$ or $s+i$; choosing the side that keeps it at most $2h-|s|$ keeps its dimension at most $p(2h)\approx p(h)^{\sqrt{2}}$.

4. Proofs of the lemmas

Proof of Lemma 5. For $b\in V_h$ and a monomial a, in $a[-h]b$ only terms of (47) with all $k_i>0$ survive: if some $k_i<0$, the positive ones add up to more than h and the differentiations kill b. Expand $\prod_i y_{\mu_i}=\sum_j\prod_i\binom{\mu_i}{j_i}t_{j_i}$ $(1\le j_i\le\mu_i)$ and apply (47) to each monomial, listing its parts as $j_1,\dots,j_n$:

$$\Big(\prod_i y_{\mu_i}\Big)[-h]\,b=\sum_{\substack{k_i\ge 1\\ \sum k_i=h}}\prod_i\vartheta_{\mu_i}(k_i)\prod_i\nabla_{k_i}b,\qquad \vartheta_n(k)=\sum_{j=1}^{n}\binom{n}{n-j}\binom{-k-1}{j-1}=\binom{n-k-1}{n-1}$$

by Vandermonde's identity (valid for all integer upper indices). For $1\le k<n$, $0\le n-k-1<n-1$ and $\vartheta_n(k)=0$; $\vartheta_n(n)=\binom{-1}{n-1}=(-1)^{n-1}$. So a nonzero term has $k_i\ge\mu_i$ for all i; as both sides add up to h, $k_i=\mu_i$. The signs cancel those in a^μ, so $a^\mu[-h]b=\prod_i\nabla_{\mu_i}b$. For $b=t^\nu$, $\prod_i\mu_i\partial_{t_{\mu_i}}t^\nu$ is zero unless $\mu\subseteq\nu$ as multisets, i.e. $\mu=\nu$, and then equals z_μ. Hence $a^\mu[-h]b^\nu=\frac{h!}{z_\nu}z_\mu\delta_{\mu\nu}=h!\,\delta_{\mu\nu}$. □

Proof of Lemma 4. All three statements are linear in a and b (since $a^{\langle i\rangle}$ is linear in a), so it suffices to treat monomials. Let $a=t^\mu$, $|\mu|=h'\le h$, $b=t^\nu$, $|\nu|=h$, and put $H=h'+h$; then $a^{\langle i\rangle}=\binom{H}{i}a$. The terms of (48) with $i>H$ vanish, so (48) is a sum over $0\le i\le H$, which is what the expansion of $(\omega_1-\omega_2)^H$ below produces.

Rings. Let $\mathcal{L}=\mathbb{Q}[\omega_1^{\pm1},\omega_2^{\pm1}]$; $\mathcal{L}^+$ the formal series $\sum_{j\ge j_0}q_j(\omega_1)\omega_2^j$ with $q_j\in\mathbb{Q}[\omega_1^{\pm1}]$, and $\mathcal{L}^-$ the same with ω_1,ω_2 exchanged. In $\mathcal{L}^\pm$, $\omega_1-\omega_2$ has the inverse $\varepsilon_+=\sum_{m\ge0}\omega_1^{-m-1}\omega_2^m$, resp. $\varepsilon_-=-\sum_{m\ge0}\omega_2^{-m-1}\omega_1^m$. From $(\omega_1-\omega_2)\varepsilon_\pm=1$: $\partial_{\omega_1}\varepsilon_\pm=-\varepsilon_\pm^2$, $\partial_{\omega_2}\varepsilon_\pm=\varepsilon_\pm^2$. Hence $\partial_{\omega_2}^{n-1}\varepsilon^2=n!\,\varepsilon^{n+1}$ and

$$\frac{1}{(j-1)!(n-1)!}\,\partial_{\omega_1}^{j-1}\partial_{\omega_2}^{n-1}\varepsilon_\pm^2=\Gamma_{j,n}\,\varepsilon_\pm^{j+n},\qquad \Gamma_{j,n}=(-1)^{j-1}\frac{(j+n-1)!}{(j-1)!\,(n-1)!},\tag{49}$$

the same number for both signs.

Generating functions. For $n\ge1$ let

$$J_n^+(\omega)=\sum_{l\ge1}\binom{l-1}{n-1}\nabla_{-l}\,\omega^{l-n},\qquad J_n^-(\omega)=\sum_{k\ge1}\binom{-k-1}{n-1}\nabla_k\,\omega^{-k-n}.$$

Reading (47) with $k_i = -l_i < 0$ for i in a set S of parts and $k_i > 0$ otherwise,

$$Y(t^\mu, \omega) := \sum_r t^\mu[r]\, \omega^{r-h'} = \sum_S \prod_{i\in S} J^+_{\mu_i}(\omega) \prod_{i\notin S} J^-_{\mu_i}(\omega),$$

creation operators to the left. Take commuting indeterminates z_n, z'_n ($n \geq 1$), $A^\pm = \sum_n z_n J^\pm_n(\omega_1)$, $B^\pm = \sum_n z'_n J^\pm_n(\omega_2)$. The J^+'s commute with each other, as do the J^-'s, so expanding the exponentials and collecting $z^\mu = \prod z_{\mu_i}$,

$$e^{A^+} e^{A^-} = \sum_\mu \frac{z^\mu}{\mu!}\, Y(t^\mu, \omega_1), \qquad \mu! = \prod_k m_k(\mu)!,$$

because the coefficient of z^μ collects the terms with $\alpha_n + \beta_n = m_n(\mu)$ factors J^+_n, J^-_n; and $1/(\alpha_n!\beta_n!) = \binom{m_n}{\alpha_n}/m_n!$, where $\binom{m_n}{\alpha_n}$ counts the ways to choose which α_n of the m_n parts equal to n lie in S. Likewise for $B^\pm$ with ω_2.

Commuting the middle factors. B^+ is multiplication by $\sum_k B_k t_k$, $B_k = \sum_n z'_n \binom{k-1}{n-1} \omega_2^{k-n}$, and $A^- = \sum_k A_k \nabla_k$, $A_k = \sum_j z_j \binom{-k-1}{j-1} \omega_1^{-k-j}$. By Taylor's formula, on the coefficient of each power of ω_2, e^{A^-} acts as the ring homomorphism $t_k \mapsto t_k + kA_k$, so $e^{A^-} e^{B^+} = e^\kappa e^{B^+} e^{A^-}$ with

$$\kappa = \sum_k k A_k B_k = \sum_{j,n} z_j z'_n \sum_{k\geq 1} k \binom{-k-1}{j-1} \binom{k-1}{n-1} \omega_1^{-k-j} \omega_2^{k-n} = \sum_{j,n} z_j z'_n\, \Gamma_{j,n}\, \varepsilon_+^{j+n} \quad \text{in } \mathcal{L}^+,$$

since the inner sum is the left side of (49) applied to $\varepsilon_+^2 = \partial_{\omega_2}\varepsilon_+ = \sum_k k\, \omega_1^{-k-1}\omega_2^{k-1}$. In the same way $e^{B^-} e^{A^+} = e^{\kappa'} e^{A^+} e^{B^-}$ in $\mathcal{L}^-$, $\kappa' = \sum z_j z'_n \Gamma_{j,n} \varepsilon_-^{j+n}$ (now using $\varepsilon_-^2 = -\partial_{\omega_1}\varepsilon_- = \sum_k k\, \omega_2^{-k-1}\omega_1^{k-1}$). With the fully normal-ordered $\mathcal{N} = e^{A^+} e^{B^+} e^{A^-} e^{B^-}$ (multiplications commute, and so do differentiations):

$$e^{A^+} e^{A^-} e^{B^+} e^{B^-} = e^\kappa \mathcal{N} \text{ in } \mathcal{L}^+, \qquad e^{B^+} e^{B^-} e^{A^+} e^{A^-} = e^{\kappa'} \mathcal{N} \text{ in } \mathcal{L}^-. \tag{51}$$

In which ring these identities hold. Each side of (51), and of the two commutation rules before it, is a formal power series in the z_n, z'_n whose coefficients are $\mathrm{End}(V)$-valued formal series in $\omega_1^{\pm 1}, \omega_2^{\pm 1}$. Such a product is read as follows: fix a monomial in the z, z', a power of ω_2 (resp. of ω_1), and two monomials $t^\sigma, t^{\sigma'}$; then $[t^\sigma](\cdots)\, t^{\sigma'}$ is a finite sum of Laurent monomials in the other variable, and an identity means that these finite sums agree for every such choice. The finiteness is the argument of [1, App. B]. In every product here e^{A^+} stands left of e^{A^-} and e^{B^+} left of e^{B^-}. If moreover e^{B^-} does not stand left of e^{A^+}, the result lies in $\mathcal{L}^+$: (i) the differentiations from B^- are applied before all multiplications, so their indices add up to at most $|\sigma'|$; (ii) the ω_2-exponent of a term $z'_n \nabla_{-k}$ of B^+ is $k - n \geq 0$ and grows with k, while those from B^- are bounded by (i), so a fixed power of ω_2 leaves finitely many choices for the multiplications from B^+; (iii) with the z-monomial and all weights now bounded, the indices from $A^\pm$ are bounded too. The ω_2-exponents are bounded below, so the coefficient lies in $\mathcal{L}^+$. Likewise the result lies in $\mathcal{L}^-$ (with ω_1, ω_2 exchanged) if e^{A^-} does not stand left of e^{B^+}. In particular κ, e^κ and both sides of the first identity in (51) are in $\mathcal{L}^+$, and κ' and the second identity in $\mathcal{L}^-$.

One Laurent polynomial. Fix $\sigma, \sigma' \in \mathcal{P}_E$ and write $\langle X \rangle = [t^\sigma] X\, t^{\sigma'}$. Every coefficient $n_{\mu'',\nu''}$ of $z^{\mu''} z'^{\nu''}$ in $\langle \mathcal{N} \rangle$ is in $\mathcal{L}$ (only ∇_k with $|k| \leq E$ contribute between two weight-E states), and $n_{\emptyset,\emptyset} = \delta_{\sigma\sigma'}$. The coefficient of $z^{\mu'} z'^{\nu'}$ in e^κ is $\Gamma_{\mu',\nu'} \varepsilon_+^{|\mu'|+|\nu'|}$ for a number $\Gamma_{\mu',\nu'}$ (each $z_j z'_n$ carries ε^{j+n}), and in $e^{\kappa'}$ it is $\Gamma_{\mu',\nu'} \varepsilon_-^{|\mu'|+|\nu'|}$. Comparing coefficients of $z^\mu z'^\nu$ in (51):

$$f_\pm := \langle Y(a,\omega_1) Y(b,\omega_2) \rangle \text{ resp. } \langle Y(b,\omega_2) Y(a,\omega_1) \rangle = \mu!\, \nu! \sum \Gamma_{\mu',\nu'}\, \varepsilon_\pm^{|\mu'|+|\nu'|}\, n_{\mu'',\nu''},$$

summed over splittings $\mu = \mu' \sqcup \mu''$, $\nu = \nu' \sqcup \nu''$. Since $|\mu'| + |\nu'| \le H$, multiplying by $(\omega_1 - \omega_2)^H$ gives in both cases the same Laurent polynomial $F = \mu!\nu! \sum \Gamma_{\mu',\nu'} (\omega_1 - \omega_2)^{H-|\mu'|-|\nu'|} n_{\mu'',\nu''} \in \mathcal{L}$, with $F(1,1) = \mu!\nu!\,\Gamma_{\mu,\nu}\,\delta_{\sigma\sigma'}$ (every other term has a factor $\omega_1 - \omega_2$).

Reading off. Only weight-preserving products contribute to $\langle\cdot\rangle$, so

$$f_+ = \sum_r \langle a[r] b[-r] \rangle\, \omega_1^{r-h'} \omega_2^{-r-h}, \qquad f_- = \sum_r \langle b[-r] a[r] \rangle\, \omega_1^{r-h'} \omega_2^{-r-h},$$

with no terms for $r > E$ in f_+ ($b[-r]t^{\sigma'} \in V_{E-r}$) and none for $r < -E$ in f_-. Multiplying by $(\omega_1 - \omega_2)^H = \sum_i (-1)^i \binom{H}{i} \omega_1^{H-i} \omega_2^i$ and collecting $\omega_1^s \omega_2^{-s}$ ($r = s - h + i$):

$$F = \sum_s F_s\, \omega_1^s \omega_2^{-s}, \qquad F_s = \langle Q_s(a,b) \rangle = \langle Q'_s(a,b) \rangle.$$

This is (a). For $s > E + h$, every $r = s - h + i \ge s - h > E$, so all terms of the first form vanish; for $s < -E - h$, every $r = s - h + i \le s - h + H = s + h' < -E$, so all terms of the second form vanish ($a[r]t^{\sigma'} \in V_{E+r} = 0$). This is (b). Finally $\sum_s F_s = F(1,1) = \mu!\nu!\Gamma_{\mu,\nu}\delta_{\sigma\sigma'}$, and $\Gamma_{\mu,\nu}$ does not depend on E. To identify it take $E = 0$: then $\langle \mathcal{N} \rangle = 1$ (the differentiations kill 1, and the constant term of $e^{A^+} e^{B^+} 1$ is 1), so $f_+ = \mu!\nu!\Gamma_{\mu,\nu}\varepsilon_+^H$, whose coefficient of ω_2^0 is $\mu!\nu!\Gamma_{\mu,\nu}\,\omega_1^{-H}$. In $f_+ = \sum_r \langle a[r]b[-r] \rangle \omega_1^{r-h'} \omega_2^{-r-h}$ the ω_2^0 term is $r = -h$, i.e. $a[-h]\,(b[h]1)\,\omega_1^{-H}$. And $b[h]1 = b$: on 1 only terms of (47) with all $k_i = -l_i < 0$ survive, $\sum l_i = h = \sum \nu_i$, and $\binom{l_i - 1}{\nu_i - 1} = 0$ for $l_i < \nu_i$, so $l_i = \nu_i$ and the term is t^ν. Hence $\mu!\nu!\Gamma_{\mu,\nu} = a[-h]b$, which is (c). □

Remark 6. For homogeneous a, (a) is the coefficient form of locality of the fields $Y(a,\omega_1), Y(b,\omega_2)$ of the Heisenberg vertex algebra, $(\omega_1 - \omega_2)^H [Y(a,\omega_1), Y(b,\omega_2)] = 0$ [3, §2.1 and Prop. 2.4], [7, Def. 1.5.1], restricted to the weight-preserving products, as [1, App. H] observes. A reader who knows that theory can take (a) for homogeneous a from it. It does not by itself give (b), (c), or the bookkeeping for inhomogeneous a through $a^{\langle i \rangle}$; those are proved above directly, and they are what the identity (1) needs.

5. Counting

Let $\kappa_{\mathrm{HR}} = \pi\sqrt{2/3}/\ln 2 \approx 3.70$. By Hardy–Ramanujan, $\kappa_{\mathrm{HR}}\sqrt{E} - \log_2 E - C \le \log_2 p(E) \le \kappa_{\mathrm{HR}}\sqrt{E} + C$ (the form used in [1, (46)]). For $N = 2^\lambda$ and h least with $p(h) \ge N$: $p(h-1) < 2^\lambda$ gives $\kappa_{\mathrm{HR}}\sqrt{h} \le \lambda + O(\log\lambda)$ and $h = O(\lambda^2)$. Since p is nondecreasing, $K_h \le (2h+1)(4h+1)\,p(2h)$, and $\log_2 p(2h) \le \sqrt{2}\,\kappa_{\mathrm{HR}}\sqrt{h} + C \le \sqrt{2}\,\lambda + O(\log\lambda)$. So $K_h \le N^{\sqrt{2}} (\log N)^{O(1)}$. Exact values (computed by `check_lemmas.py`):

λ	h	$\log_2 K_h$	$\log_2 K_h/\lambda$
10	23	22.7	2.272
20	61	38.4	1.922
40	186	68.9	1.723
60	372	98.5	1.641
70	489	113.2	1.618
80	620	127.8	1.598
100	927	156.9	1.569
200	3361	300.9	1.505
400	12663	586.4	1.466
600	27849	870.9	1.451

The exponent first drops below 1.6147 at $\lambda = 72$, and approaches $\sqrt{2}$ only like $\log\lambda/\lambda$. For small N the identity uses far more than N^2 products ($K_4 = 307$ for $N \le 5$, $K_5 = 707$ for $N \le 7$): it is purely asymptotic.

6. What is checked, and where

Step	Lean declaration (paths relative to `eth/proof/rates/GeneralPieces/`; those beginning `statements/` or `rates/` are relative to `eth/proof/`)
Definition 7, Theorem B (statements)	`Pieces.Family.Is` (`statements/PiecesStatement.lean`); `AppHalfPieces.ThmHalfPiecesPure`, `LemmaQ`, `LemmaDual`, `PieceFacts` (`statements/AppHalfPiecesStatement.lean`)
(47): finiteness, linearity; (O1)	`GeneralPieces.opFacts` (`Ring/OpFacts.lean`); `GeneralPieces.o1` (`Ring/Op.lean`)
(51), the two orders	`GeneralPieces.GenF.stTwoOrders` (`GenF/Commute.lean`); `GeneralPieces.GenF.stExpansion` (`GenF/Main.lean`)
One Laurent polynomial F; $F(1,1)$; the constant	`Expansion.dW_pow_mul_fP`, `Expansion.dW_pow_mul_fM`, `Expansion.ev_F` (`Series/OneLaurent.lean`); `Expansion.const` (`Series/Constant.lean`), all in `GeneralPieces.Series`
Lemma 4 (a), (b), (c)	`GeneralPieces.Series.Qk_eq_Qk'_of`, `Qk_eq_zero_of`, `sum_Qk_of` (`Series/LemmaQ.lean`); assembled as `GeneralPieces.lemmaQ` (`Ring/LemmaQ.lean`) via `GeneralPieces.stBridge` (`Ring/Bridge.lean`)
Lemma 5; $h!/z_\mu \in \mathbb{Z}$	`GeneralPieces.Dual.lemmaDual`, with `theta_eq`, `theta_of_lt`, `theta_self`; `GeneralPieces.Dual.zDvd` (`Ring/Dual.lean`)
(1), (54), (55), integrality	`GeneralPieces.pieceFacts` (`Ring/Pieces.lean`); per piece `GeneralPieces.ThmHalfPieces.sum_A`, `row_A`, `col_A` (`Ring/ThmHalfPieces.lean`)
Counting, $h = O(\lambda^2)$	`GeneralPieces.Bits.exists_bits_le`, `exists_hOf_le` (`Ring/Bits.lean`), which take (46) as a hypothesis; (46), with $B \log_2 E$ in place of $\log_2 E$, is `ClauseBound.Part.exists_bounds` (`rates/ClauseBound/Part/Bounds.lean`), used in `thmHalfPiecesPure`
Theorem B	`GeneralPieces.thmHalfPiecesPure` (`Ring/ThmHalfPieces.lean`), applied to `AppHalfPieces.pieceFacts` (`Main.lean`)
Corollary 2, bound form over $\mathbb{Q}$, $N = 2^\lambda$	`hurwitzSqrt2` (our `HurwitzProof.lean`)

Independent numerical checks. `check_lemmas.py` (Python, exact integer arithmetic, written from the paper's text, not from [8] or the Lean code) implements (47), (48), a^μ, b^μ and checks: the Heisenberg relations; Lemma 5 for all $\mu, \nu \in \mathcal{P}_h$, $h \le 5$; Lemma 4(a), (b), (c) for $a = a^\mu$ (inhomogeneous), $b = b^\nu$, $h \le 4$, on every basis monomial of V_E, $E \le 4$, for $|s| \le 2(E+h)+2$ (for $s > E+h$ the vanishing of Q_s is immediate, but for $s < -(E+h)$, and for Q'_s with $s > E+h$, it is a real cancellation; the lemma for general a, b is not tested numerically); (1) on all $p(h)^4$ index tuples for $h \le 5$; both factorizations term by term for $h \le 4$; and the identity of Theorem 1, multiplied out as a polynomial in $2p(h)$ variables, for $h = 2, \dots, 5$ ($K_h = 40, 120, 307, 707$). All 44 checks pass (about 160 seconds on a laptop). [8] reports the same identity checked for $N = 4$ and $N = 8$ with a different implementation; we reproduced its $N = 4$ case here.

Credit and provenance. The observation that Theorem B of [1] gives $N^{\sqrt{2}+o(1)}$ for Hurwitz's problem was published first by J. P. Siemaszko on 7 November 2026 [8], including the translation to [6, Lemma 3.5] and instance checks. The construction and the proofs above are those of [1], represented. This note, its Lean bridge and its scripts were produced with Claude models (Anthropic) in sessions directed by the author, who has not verified the mathematics independently of the checks described.

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